Chapter 9: Graphs

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 9 (Graphs) of Data Structures.

Technical / Interview Questions

1. Define a graph. How is it different from a tree?
2. Explain the difference between directed and undirected graphs with examples.
3. What is the difference between adjacency matrix and adjacency list?
4. When would you prefer adjacency list over adjacency matrix?
5. Define degree, in-degree, and out-degree.
6. State and prove the Handshaking Lemma.
7. What is a complete graph? How many edges does it contain?
8. Define sparse and dense graphs.
9. What is a biconnected graph? What is an articulation point?
10. Explain the working principle of DFS.
11. Explain the difference between BFS and DFS.
12. What is a spanning tree? How many edges does it contain?
13. Explain Dijkstra's algorithm and its limitations.
14. Why does Dijkstra's algorithm fail for negative weights?
15. Explain Bellman–Ford algorithm and its advantages.
16. Compare Bellman–Ford and Dijkstra algorithms.
17. Explain Floyd–Warshall algorithm.
18. What is a Minimum Spanning Tree?
19. Differentiate between Prim's and Kruskal's algorithms.
20. What is Topological Sorting? When is it applicable?
21. Explain Kahn's Algorithm.
22. What is a Directed Acyclic Graph (DAG)?
23. What are strongly connected components?
24. Explain Graph ADT and its operations.
25. What are real-world applications of graphs?
26. How can cycles be detected in a graph?
27. What is the time complexity of BFS and DFS?
28. Explain articulation point detection using DFS.
29. What is the difference between simple graph and multigraph?
30. What is a block-cut tree?

Programming Exercises

1. Write a program to represent a graph using adjacency matrix.
2. Write a program to represent a graph using adjacency list.
3. Implement DFS traversal.
4. Implement BFS traversal.
5. Write a program to detect cycle in an undirected graph.
6. Write a program to detect cycle in a directed graph.

7. Implement Dijkstra's shortest path algorithm.
8. Implement Bellman–Ford algorithm.
9. Implement Floyd–Warshall algorithm.
10. Write a program to find Minimum Spanning Tree using Prim's algorithm.
11. Write a program to find Minimum Spanning Tree using Kruskal's algo rithm.
12. Write a program to perform Topological Sort.
13. Write a program to check whether a graph is connected.
14. Write a program to find articulation points.
15. Write a program to find biconnected components.
16. Write a program to count number of leaf vertices in a graph.
17. Write a program to check whether a graph is bipartite.
18. Write a program to compute in-degree and out-degree of all vertices.
19. Write a program to find number of connected components.
20. Write a menu-driven program implementing Graph ADT operations.

Multiple Choice Questions

1. A graph with no cycles is called:

(a) Complete Graph
(b) Cyclic Graph
(c) Acyclic Graph
(d) Regular Graph

2. In an undirected graph, the sum of degrees of all vertices is equal to:

(a) e
(b) 2e
(c) V
(d) V^2

3. Which data structure is used in BFS?

(a) Stack
(b) Queue
(c) Tree
(d) Heap

4. Which traversal technique uses backtracking?

(a) BFS
(b) DFS
(c) Prim's
(d) Kruskal's

5. Dijkstra's algorithm cannot handle:

(a) Weighted Graph
(b) Directed Graph
(c) Negative Edge Weights
(d) Sparse Graph

6. Time complexity of BFS is:

(a) O(V)
(b) O(E)
(c) O(V + E)
(d) $O(V^2)$

7. A tree with n vertices has how many edges?

(a) n
(b) $n-1$
(c) $n+1$
(d) n^2

8. Floyd–Warshall algorithm is used for:

(a) MST
(b) Single-source shortest path
(c) All-pairs shortest path
(d) Traversal

9. A complete graph with n vertices has how many edges (undirected)?

(a) $n(n-1)$
(b) n^2
(c) $n(n-1)/2$
(d) n

10. Topological sorting is applicable only to:

(a) Undirected Graph
(b) Cyclic Graph
(c) Directed Acyclic Graph
(d) Tree

11. An articulation point is:

(a) A vertex with degree 1
(b) A vertex whose removal increases components
(c) A vertex with maximum degree
(d) A leaf vertex

12. Prim's algorithm is used to find:

(a) Shortest Path
(b) Spanning Tree
(c) MST
(d) Cycle

13. Kruskal's algorithm uses which technique?

(a) Dynamic Programming
(b) Greedy
(c) Divide and Conquer
(d) Backtracking

14. In-degree is applicable to:

(a) Undirected Graph
(b) Directed Graph
(c) Tree
(d) Forest

15. Which representation is better for sparse graphs?

(a) Adjacency Matrix
(b) Adjacency List
(c) Incidence Matrix
(d) None

16. A bipartite graph can be colored using:

(a) 1 color
(b) 2 colors
(c) 3 colors
(d) 4 colors

17. The maximum number of edges in a simple directed graph with V vertices is:

(a) $V(V-1)$
(b) V^2
(c) $V(V-1)/2$
(d) V

18. DFS time complexity is:

(a) $O(V)$
(b) $O(E)$
(c) $O(V + E)$
(d) $O(V^2)$

19. A connected acyclic undirected graph is:

(a) Forest
(b) Tree
(c) DAG
(d) Multigraph

20. Bellman–Ford can detect:

(a) MST
(b) Negative Cycle
(c) Strongly Connected Component
(d) Articulation Point

21. A graph with no edges is called:

(a) Null Graph
(b) Complete Graph
(c) Dense Graph
(d) Regular Graph

22. Which algorithm is used for topological sorting?

(a) DFS
(b) Kahn's Algorithm
(c) Both A and B
(d) Dijkstra

23. The adjacency matrix of an undirected graph is:

(a) Diagonal
(b) Symmetric
(c) Sparse
(d) Rectangular

24. A spanning tree of a graph with V vertices has:

(a) V edges
(b) $V-1$ edges
(c) E edges
(d) V^2 edges

25. Which algorithm is best suited for dense graphs for all-pairs shortest path?

(a) Dijkstra
(b) Bellman–Ford
(c) Floyd–Warshall
(d) Kruskal

26. Removing an articulation point will:

(a) Reduce edges
(b) Increase components
(c) Decrease vertices
(d) Form cycle

27. Which data structure is used in Kruskal's algorithm?

(a) Stack
(b) Queue
(c) Disjoint Set
(d) Heap

28. DAG stands for:

(a) Directed Acyclic Graph
(b) Dense Acyclic Graph
(c) Directed Adjacent Graph
(d) Dual Acyclic Graph

29. Which traversal guarantees shortest path in unweighted graph?

(a) DFS
(b) BFS
(c) Prim's
(d) Kruskal's

30. The complexity of Floyd–Warshall algorithm is:

(a) $O(V^2)$
(b) $O(E \log V)$
(c) $O(V^3)$
(d) $O(V + E)$

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	B	4	B	5	C
6	C	7	B	8	C	9	C	10	C
11	B	12	C	13	B	14	B	15	B
16	B	17	A	18	C	19	B	20	B
21	A	22	C	23	B	24	B	25	C
26	B	27	C	28	A	29	B	30	C